

Proof. Given $\{f_i\}_{i \in \mathbb{N}_0}$ in $\mathcal{H}_K$, one checks that

$$K(\cdot, y) = K_y(\cdot) = \sum_{i \in \mathbb{N}_0} \langle f_i, K_y \rangle f_i(\cdot) = \sum_{i \in \mathbb{N}_0} f_i(y) f_i(\cdot)$$

which is (2.14). $\square$

Question 2.12. Find the spectrum of T_μ as μ varies in $\text{Reg}(K)$. Actually it is the selfadjoint operator $T_\mu T_\mu^* : L^2(\mu) \rightarrow L^2(\mu)$, whose spectrum we want.

It is important to keep in mind that we get a family of operators T_μ indexed by $\mu \in \text{Reg}(K)$. They are densely defined operators in the Hilbert space $\mathcal{H}_K$, but the spectra may vary with μ .

Fix a p.d. kernel $X \times X \xrightarrow{K} \mathbb{R}$ on a set X . Assign the Borel σ -algebra $\mathcal{B}_K$ and measure μ , and get $(X, \mathcal{B}_K)$ as above. What is the interconnection between the following three conditions?

- (1) $K(\cdot, x) \in L^2(\mu)$, for all $x \in X$;
- (2) $\{f_i\} \subset L^2(\mu)$, for all ONB $\{f_i\}$ in $\mathcal{H}_K$;
- (3) How is the system $\{f_i\}$ related to the spectral properties of $T_\mu T_\mu^* : L^2(\mu) \rightarrow L^2(\mu)$?

A related question: Consider a p.d. kernel K on $X \times X$. Suppose X is discrete and countable, equipped with the counting measure γ on X , i.e.,

$$\gamma(B) = \#(B), \quad \forall B \subset X. \quad (2.16)$$

When is

$$\gamma \in \text{Reg}(K)? \quad (2.17)$$

We shall give a precise solution to (2.17) in the following section.

3. OPERATORS FROM P.D. KERNELS, AND SOME OF THEIR APPLICATIONS

Here, we consider a special case in the setting of Section 2, where a p.d. kernel admits a factorization in some $L^2(\mu)$ -space with $\mu \in \text{Reg}(K)$.

Let $X \times X \xrightarrow{K} \mathbb{R}$ be a p.d. kernel on a set X . Let $\mathcal{H}_K$ be the associated RKHS. Assume:

- (1) $\mu \in \text{Reg}(K)$;
- (2) $\{K_x\}_{x \in X}$ is dense in $L^2(\mu)$;
- (3) for all $F \in \mathcal{H}_K$,

$$\|F\|_{\mathcal{H}_K} \geq \|F\|_{L^2(\mu)}.$$

Lemma 3.1. Let $(\mathcal{H}_j, \|\cdot\|_j)$, $j = 1, 2$, be a pair of Hilbert spaces. Suppose the inclusion map $J : \mathcal{H}_1 \hookrightarrow \mathcal{H}_2$ has dense image, and $\|x\|_1 \geq \|x\|_2$, for all $x \in \mathcal{H}_1$. Then there exists a unique positive (selfadjoint) operator $A \geq 1$, such that $\mathcal{H}_1 = \text{dom}(A^{1/2})$ and

$$\langle x, y \rangle_1 = \langle A^{1/2}x, A^{1/2}y \rangle_2$$

for all $x, y \in \mathcal{H}_1$.

Proof. It can be verified that

$$A := (JJ^*)^{-1} \quad (3.1)$$

is the unique positive operator having the desired properties. For more details, see e.g., [Nel69, DS88, AG93, RS75] and [JT15]. $\square$